STFEnc: A Novel Deep Encoder for Brain-Computer Interface Based on Interpretable Brain Features

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***Abstract*—Brain-computer interface (BCI) is a technology that enables direct connection and interaction of brain activity with external devices or systems. The encoding and decoding of neural signals play a crucial role in BCIs. The quality of such encodings are the key to robust and accurate information exchange and control between the brain and external devices. Currently, the limited capabilities of conventional brain signal processing is restricting a wider application of BCIs. In this paper, we propose a deep network encoder, Spatio-temporal frequency domain feature fusion encoder(STFEnc), to robustly and comprehensively encode the original electroencephalogram (EEG) data. The design of STFEnc is based on an interpretable understanding of the brain's basic structural and connectivity features. The various submodules in STFEnc were designed to ensure the different features were taken into account. We evaluated encoder performance on three motion imagery datasets and one picture stimulus dataset. The results show that our encoder performs better than traditional deep encoders and advanced deep neural network models that have excelled in extracting EEG features for classification in recent years. The code is available at https://github.com/gwkuqgfkqe/Spatio-temporal-frequency-domain-feature-fusion-encoder.**

***Index Terms*—component, formatting, style, styling, insert**

I. Introduction

In recent years, rapid advances in artificial intelligence (AI) and neuroscience have spurred significant research into integrating these fields, with brain-computer interfaces (BCIs) being a major beneficiary of this trend. BCIs enable communication between the human brain and external devices by interpreting brain signals, particularly electroencephalography (EEG) data. This integration has substantially improved the performance of many BCI applications, leading to breakthroughs such as decoding EEG signals into speech [1], [2], generating images [3], [4], and controlling robotic arms via EEG-based signals [5], [6].

A key element in these applications is the encoding of EEG signals, which involves transforming the raw neural data into a format that can be accurately interpreted by machines. This encoding ensures precise correspondence between a user's neural intent and the external action performed by the system. For example, accurate encoding models can distinguish between various brain activities, enabling the system to respond to specific intentions or commands. Figure **??** illustrates the separability achieved in our model when applied to EEG data, where different classes of neural activity are clearly distinguishable after encoding.

The human brain's structure significantly influences the content of EEG signals, and any robust BCI system must account for this. Specifically, three key characteristics of the brain affect EEG signals:

- Connectomic Information – Brain regions are interconnected through complex networks, known as the connectome. These connections play a crucial role in brain function, affecting learning, memory, and neurological disorders [7], [8]. Variations in these connections influence the structure of EEG signals.
- Hierarchical Structure – The brain is organized hierarchically, with different regions processing various levels of information, particularly in areas such as vision and motor control [9]. This hierarchy is reflected in the characteristics of EEG signals and must be considered in encoding models [10].
- Functional Network Information – EEG signals capture data on functional brain networks, including metrics like connectivity and degree distribution [11]. These network structures help reveal cognitive states and influence the overall informational content of the EEG data.

Given these factors, we propose a unified EEG encoding framework that takes into account the connectomic, hierarchical, and network properties of the brain. Our framework is designed to be versatile, allowing it to encode EEG signals for a range of tasks, such as motor imagery, visual signal decoding, and classification.

Our key contributions in this work are:

- We propose an encoding model that comprehensively captures the latent information within EEG signals, enabling its application to visual and motor tasks.
- We design a deep learning architecture that incorporates the key structural and functional characteristics of the

brain, optimizing it for diverse BCI applications.
- We rigorously evaluate the proposed framework on multiple motor imagery datasets and an image stimulus dataset to demonstrate its robustness and effectiveness across different use cases.
- We will open source our code to reproduce the results.

This approach not only enhances the quality of signal decoding but also improves the overall reliability and performance of BCI systems. By unifying different embedding models under a single, general framework, we ensure that the encoding process is both flexible and precise, making it suitable for a wide array of downstream applications.

II. Related Work

BCIs have evolved since Hans Berger's first EEG recordings [12], leading to applications like the P300 speller [13]. Modern BCIs use EEG signals for device control, often through motor imagery [14]. Early BCIs focused on classification, but recent work emphasizes encoding EEG signals for flexible downstream tasks [15], [16]. Deep learning's ability to handle multimodal, high-dimensional data has made it a popular choice for EEG analysis [17]. This flexibility enables advanced encoding strategies across time, frequency, and space domains [18].

Decoding—reconstructing stimuli from brain signals—has also advanced through deep learning, particularly with generative models like diffusion models [19], [20]. These approaches enhance BCI performance, driving applications in "mind-reading" technologies [21].

III. Methods

A. Dataset And Data Preprocessing

We evaluated our proposed encoder on four datasets, including three motor imagery datasets and one object recognition dataset. Each dataset was preprocessed to extract relevant EEG signal segments, which were then fed into our encoder for further analysis:

- BNCI 2014-001 Motor Imagery Dataset: This dataset includes EEG data from 9 subjects performing four motor imagery tasks: imagining movement of the left hand, right hand, both feet, and tongue. We split the dataset evenly, using 50% for training and 50% for testing. Each epoch contains raw, unfiltered EEG data spanning 4.5 seconds, recorded at a 250 Hz sampling rate, with 1125 samples per trial.
- High-Gamma Dataset: This dataset features EEG data recorded using 128 electrodes, later reduced to 44 sensors covering the motor cortex, across 14 subjects. The task includes four classes: left hand, right hand, both feet, and rest. We used unfiltered data with 4.5-second epochs for input to the network, with a total of 2250 samples per trial.
- BNCI 2015-001 Motor Imagery Dataset: In this dataset, subjects perform motor imagery tasks involving sustained right hand versus both feet movement. As with the previous datasets, we split it 50/50 for training and testing, using epochs of 5.5 seconds with 2816 samples per trial.
- Object Recognition Dataset: This dataset consists of six classes (human body, human face, animal body, animal face, fruit/vegetable, and inanimate objects), each shown to 10 participants. EEG signals were recorded using a 128-channel device. After preprocessing, we had 4000 trials for training and 1184 for testing, with each trial spanning 0.5 seconds.

B. Encoder

The proposed deep encoder, STFEnc, is composed of four key blocks designed to extract critical features from EEG data by leveraging brain structure and functional connectivity. Each block contributes to encoding different aspects of the data, ensuring that the full range of information is captured for downstream tasks.

- Frequency Attention (FA) Block: The FA block enhances feature representation in the frequency domain by applying attention mechanisms. First, the input EEG data is transformed to the frequency domain using the fast Fourier transform (FFT). The attention mechanism then computes weights to emphasize the most informative frequency components. These weights are applied to the original frequency data, and finally, the signal is transformed back to the time domain using inverse FFT (iFFT). This process allows the network to focus on frequency components critical for specific cognitive or motor activities.
- Graph Convolution (GC) Block: The GC block captures connectomic information by constructing a graph-based representation of the EEG channels. It creates an adjacency matrix based on the Pearson correlation between EEG channels, treating the channels as nodes and their correlations as edges. The graph convolutional network (GCN) then learns an importance matrix, which weights the connections between nodes (EEG channels), making the adjacency matrix adaptive during training. This block is critical for extracting spatial relationships and functional connectivity information from the EEG data, linking it to the brain's connectomic structure.
- Convolution (CV) Block: The convolution block encodes spatiotemporal features through sequential convolutions, reflecting the hierarchical organization of brain signals. It applies a convolution kernel across EEG channels to capture inter-channel relationships, followed by a temporal convolution that extracts sequential features from the EEG signals. This block includes separable convolutions (a combination of depthwise and pointwise convolutions) to efficiently encode both local and global spatiotemporal patterns in the data.
- ATCNet Block: The ATCNet block focuses on encoding temporal dependencies in the EEG signals, particularly critical for motor imagery tasks. By removing the classification and Softmax layers from the original ATCNet

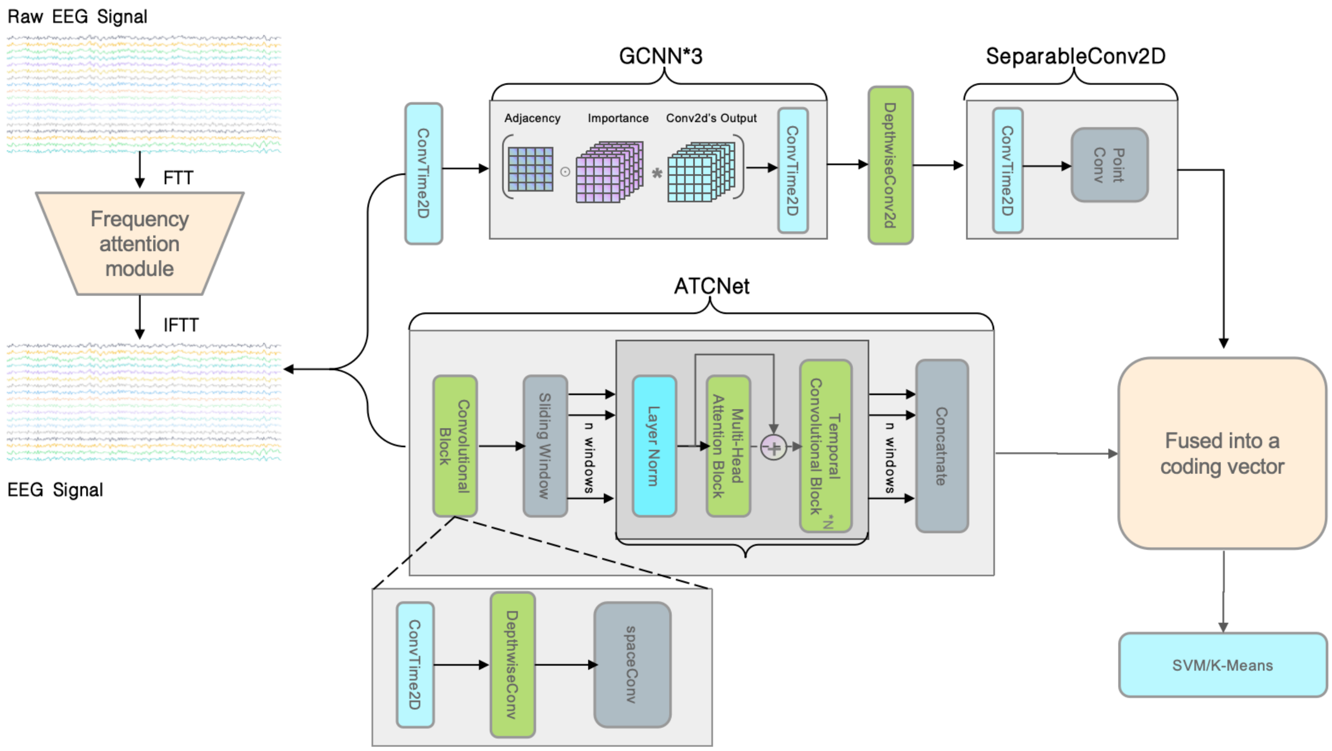

Fig. 1: Full setup of STFEnc. The various consisting submodules are explained in detail in the text.

design, we adapted this block to output an encoding vector, which is then combined with the output from the other blocks. This ensures that the final encoded vector contains rich temporal information essential for understanding the progression of neural activity over time.

The combined output of these blocks forms an encoded vector for downstream tasks. The FA block processes the input in the frequency domain, applying attention, and the GC block builds a graph-based representation of brain connectivity. The convolution and ATCNet blocks then encode spatiotemporal information before producing the final encoding vector.

C. Experiments

We tested the encoder's performance on motor imagery and object recognition datasets, comparing its classification accuracy and clustering quality against baselines. Metrics used included accuracy (ACC) and the Calinski-Harabasz (CH) score for clustering quality. Our method outperformed state-of-the-art models like ATCNet [22], EEGNet [23], and EEG-ITNet [24] on all datasets, confirming the robustness of STFEnc.

IV. Results and Discussions

A. Comparison with Baseline Models

In Table I we show the averaged and SVM input accuracies of EEG data encoded by the proposed STFEnc encoder using BNCI 2014-001, High-gamma, BNCI 2015-001 and Object datasets. We performed baseline comparisons with EEGNet, EEGITNet, ATCNet, EEGConformer, LSTM, Mul-LSTM, and CNN; some of these models are deep encoders and some are deep network models that perform well in classification tasks. The results of the replicated models are based on the hyperparameters identified in the original article, while the preprocessing, training, and evaluation follow the same procedures defined in this article. Table I shows that in all the four data sets, STFEnc performs better than all the baseline models, with a significant lead, which indicates that the basis of our hypothesis on the important contributory factors to EEG information content is sound. The average confusion matrix of STFEnc is shown in Figure **??**.

B. TSNE dimension reduction and K-means clustering

Table II summarizes of the CH scores for STFEnc as compared to the baselines. This experiment was performed on all considered datasets pairs. The CH scores were obtained by first encoding the EEG data and performing TSNE dimensionality reduction, followed by a KMeans procedure. As mentioned above, a higher CH score implied a higher-quality clustering, which is essential for high-quality reconstructions using the computed embeddings. Again, we compared with all our reproduced baselines, the results for which are based on the hyperparameters identified in the original articles, while the preprocessing, training, and evaluation follow the same procedures defined in this article. Table II shows that STFEnc scored a lower CH than EEGConformer on the BNCI 2015-001 data set, but otherwise builds a large lead over the rest

TABLE I: The accuracy of data encoding by different encoders and STFEnc and classification by SVM are compared in the table

Dataset	Subject	STFEnc	ATCNet	EEGNet	EEGITNet	EEGConformer	CNN	LSTM	Mul-LSTM
BNCI 2014-001	Avg	**79.51%**	74.68%	71.24%	73.64%	58.85%	41.68%	37.60%	40.91%
High-gamma	Avg	**91.88%**	88.75%	82.00%	82.50%	82.75%	59.75%	55.25%	56.87%
BNCI 2015-001	Avg	**81.00%**	77.04%	79.33%	65.95%	75.66%	56.75%	56.45%	56.00%
Object	Avg	**70.71%**	64.74%	64.40%	59.51%	52.20%	49.08%	39.05%	39.13%

TABLE II: The Calinski-Harabasz scores obtained after data encoding by different encoders and STFEnc, t-SNE dimensionality reduction and K-means clustering are compared in the table

Dataset	STFEnc	ATCNet	EEGNet	EEGITNet	EEGConformer	CNN	LSTM	Mul-LSTM
BNCI 2014-001	**372**	351	313	312	338	295	280	329
High-gamma	**319**	306	298	269	191	256	250	253
BNCI 2015-001	1113	1046	844	780	**1342**	457	359	321
Object	**2104**	**2104**	1731	1945	1680	1471	1361	1514

TABLE III: Ablation analysis of STFEnc was performed on four datasets

Dataset	Not Removed		Removed FA		Removed CV		Removed GC		Removed ATCNet	
	SVM	K-means	SVM	K-means	SVM	K-means	SVM	K-means	SVM	K-means
BNCI 2014-001	**79.51%**	**372**	77.51%	342	79.00%	352	78.01%	355	74.31%	264
High-gamma	**91.88%**	**319**	89.06%	304	90.64%	310	88.01%	288	84.72%	355
BNCI 2015-001	**81.00%**	**1113**	79.22%	1046	80.72%	1099	80.55%	1108	77.49%	784
Object	**70.71%**	**2104**	69.32%	1966	69.59%	2088	68.04%	1842	66.23%	1465

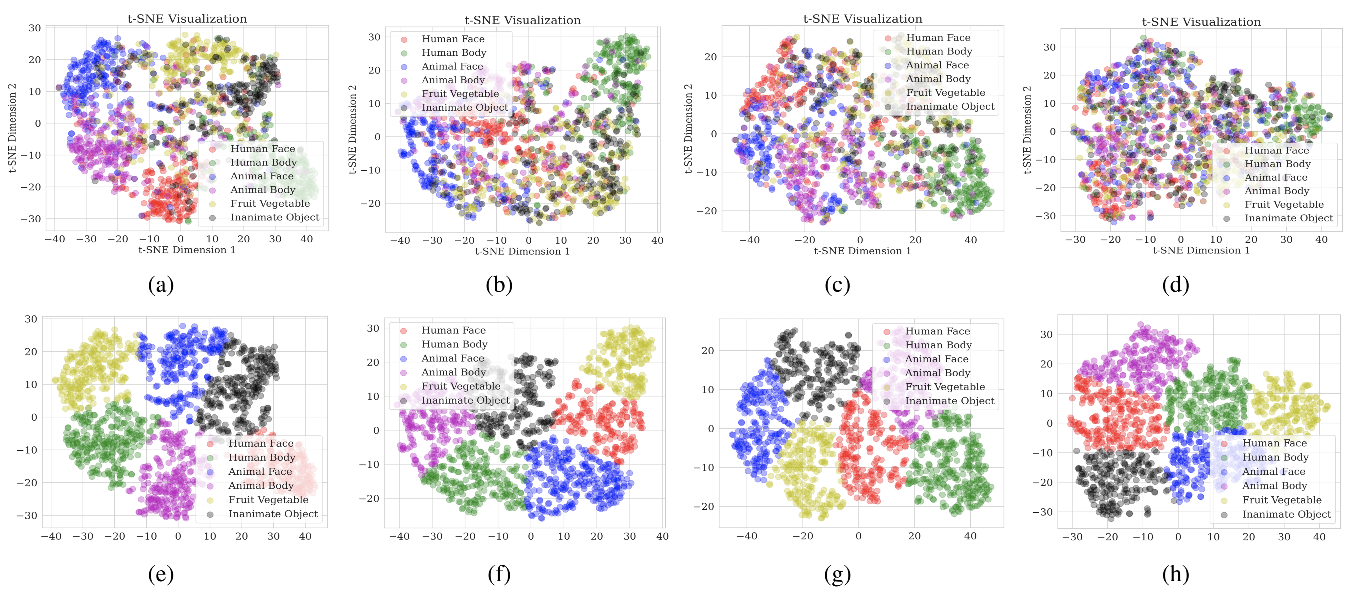

Fig. 2: The first row, from left to right, shows TSNE results from STFEnc, EEGITnet, EEGNet and MUL-LSTM on the Object dataset. The next row, from left to right, shows additional K-means visualization on the TSNE figures.

of the baseline models, which again validates our design principles. A full visualization of the TSNE and k-Means results are shown in Figure 2. From the four visual datasets in use in this work, we selected the Object dataset for full visualization. Evaluation on all models were done and the corresponding results plotted.

C. Ablation analysis

To evaluate the importance of each component, we performed ablation studies by removing individual blocks. The results show that removing the ATCNet block caused the most significant performance drop, indicating the critical role of temporal features in EEG signal encoding.

V. Results and Discussions

Our model achieves superior classification accuracy across all datasets (Table I). Clustering analysis using CH scores (Table II) further supports the high-quality separation of encoded EEG data. Ablation studies (Table III) confirmed that all components contribute meaningfully to performance, especially the ATCNet and GC blocks. Visualizations using TSNE and K-means clustering demonstrate clear separability between different classes of EEG data (Figure 2).

VI. Conclusion

We proposed STFEnc, an encoder for EEG signals designed to capture frequency, spatiotemporal, and connectomic information. This model demonstrated state-of-the-art performance across various datasets and tasks, confirming the importance of incorporating brain structure and connectivity into EEG encoding models. Future work will extend the framework to other BCI applications and further optimize model efficiency.

References

[1] X. Chen, R. Wang, A. Khalilian-Gourtani, L. Yu, P. Dugan, D. Friedman, W. Doyle, O. Devinsky, Y. Wang, and A. Flinker, "A neural speech decoding framework leveraging deep learning and speech synthesis," *Nature Machine Intelligence*, vol. 6, no. 4, pp. 467–480, apr 2024. [Online]. Available: https://doi.org/10.1038/s42256-024-00824-8

[2] G. K. Anumanchipalli, J. Chartier, and E. F. Chang, "Speech synthesis from neural decoding of spoken sentences," *Nature*, vol. 568, no. 7753, pp. 493–498, apr 2019. [Online]. Available: https://doi.org/10.1038/s41586-019-1119-1

[3] N. Koide-Majima, S. Nishimoto, and K. Majima, "Mental image reconstruction from human brain activity: Neural decoding of mental imagery via deep neural network-based bayesian estimation," *Neural Networks*, vol. 170, pp. 349–363, 2024. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0893608023006470

[4] Y. Song, B. Liu, X. Li, N. Shi, Y. Wang, and X. Gao, "Decoding natural images from eeg for object recognition," 2024.

[5] M. Kim, M. S. Choi, G. R. Jang, J. H. Bae, and H. S. Park, "Eeg-controlled tele-grasping for undefined objects," *Frontiers in Neurorobotics*, vol. 17, p. 1293878, dec 2023.

[6] J. Meng, S. Zhang, A. Bekyo, J. Olsoe, B. Baxter, and B. He, "Noninvasive electroencephalogram based control of a robotic arm for reach and grasp tasks," *Scientific Reports*, vol. 6, p. 38565, dec 2016. [Online]. Available: https://doi.org/10.1038/srep38565

[7] H. Gast and Y. Assaf, "Weighting the structural connectome: Exploring its impact on network properties and predicting cognitive performance in the human brain," *Network Neuroscience*, vol. 8, no. 1, pp. 119–137, 04 2024. [Online]. Available: https://doi.org/10.1162/netn_a_00342

[8] S. H. Bennett, A. J. Kirby, and G. T. Finnerty, "Rewiring the connectome: Evidence and effects," *Neuroscience Biobehavioral Reviews*, vol. 88, pp. 51–62, May 2018, epub 2018 Mar 11.

[9] T. Serre, *Hierarchical Models of the Visual System*. New York, NY: Springer New York, 2013, pp. 1–12. [Online]. Available: https://doi.org/10.1007/978-1-4614-7320-6_345-1

[10] G. K. Steinke and R. F. Galán, "Brain rhythms reveal a hierarchical network organization," *PLOS Computational Biology*, vol. 7, no. 10, pp. 1–15, 10 2011. [Online]. Available: https://doi.org/10.1371/journal.pcbi.1002207

[11] G. Chiarion, L. Sparacino, Y. Antonacci, L. Faes, and L. Mesin, "Connectivity analysis in eeg data: A tutorial review of the state of the art and emerging trends," *Bioengineering*, vol. 10, no. 3, 2023. [Online]. Available: https://www.mdpi.com/2306-5354/10/3/372

[12] H. Berger, "Über das elektrenkephalogramm des menschen," *Archiv für Psychiatrie und Nervenkrankheiten*, vol. 87, no. 1, pp. 527–570, 1929.

[13] L. A. Farwell and E. Donchin, "Talking off the top of your head: Toward a mental prosthesis utilizing event-related brain potentials," *Electroencephalography and Clinical Neurophysiology*, vol. 70, no. 6, pp. 510–523, 1988.

[14] G. Pfurtscheller, D. Flotzinger, and J. Kalcher, "Brain-computer interface: a new communication device for handicapped persons," *Journal of Microcomputer Applications*, vol. 16, pp. 293–299, 1993. [Online]. Available: https://api.semanticscholar.org/CorpusID:62582338

[15] E. Hernandez and J. Andreas, "The low-dimensional linear geometry of contextualized word representations," *arXiv preprint arXiv:2105.07109*, 2021.

[16] L. Ward, A. Agrawal, A. Choudhary, and C. Wolverton, "A general-purpose machine learning framework for predicting properties of inorganic materials," *npj Computational Materials*, vol. 2, no. 1, pp. 1–7, 2016.

[17] D. Lahat, T. Adali, and C. Jutten, "Multimodal data fusion: an overview of methods, challenges, and prospects," *Proceedings of the IEEE*, vol. 103, no. 9, pp. 1449–1477, 2015.

[18] B. Ghoraani and S. Krishnan, "Time–frequency matrix feature extraction and classification of environmental audio signals," *IEEE transactions on audio, speech, and language processing*, vol. 19, no. 7, pp. 2197–2209, 2011.

[19] Q. Zhou, C. Du, D. Li, H. Wang, J. K. Liu, and H. He, "Neural encoding and decoding with a flow-based invertible generative model," *IEEE Transactions on Cognitive and Developmental Systems*, 2022.

[20] Y. Takagi and S. Nishimoto, "High-resolution image reconstruction with latent diffusion models from human brain activity," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 14 453–14 463.

[21] A. Realo, J. Allik, A. Nõlvak, R. Valk, T. Ruus, M. Schmidt, and T. Eilola, "Mind-reading ability: Beliefs and performance," *Journal of Research in Personality*, vol. 37, no. 5, pp. 420–445, 2003. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0092656603000217

[22] H. Altaheri, G. Muhammad, and M. Alsulaiman, "Physics-informed attention temporal convolutional network for eeg-based motor imagery classification," *IEEE transactions on industrial informatics*, vol. 19, no. 2, pp. 2249–2258, 2022.

[23] V. J. Lawhern, A. J. Solon, N. R. Waytowich, S. M. Gordon, C. P. Hung, and B. J. Lance, "Eegnet: a compact convolutional neural network for eeg-based brain–computer interfaces," *Journal of neural engineering*, vol. 15, no. 5, p. 056013, 2018.

[24] A. Salami, J. Andreu-Perez, and H. Gillmeister, "Eeg-itnet: An explainable inception temporal convolutional network for motor imagery classification," *IEEE Access*, vol. 10, pp. 36 672–36 685, 2022.